# Practice workbook · Market failure, public goods and free riders

Market failure as an inefficient allocation of resources, its causes (poor information, lack of competition, public goods), non-excludability and non-rivalry, the free rider problem, and why government intervenes or provides.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: ECO-T11-18

# 1. Question 4 · part b · 6 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part b only.

**4** A restaurant in Cyprus can seat 136 diners each evening. The demand for seats from diners over three evenings is shown in Figure 5.

| Evening | Quantity of seats demanded |
|---|---|
| Thursday | 98 |
| Friday | 114 |
| Saturday | 107 |

**Figure 5**

(a) Calculate the **excess supply** of seats if the quantity supplied during these three evenings totalled 408. You are advised to show your working.

**(2)**

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# 1. Question 4 · part b · 6 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part b only.

Street lighting is provided by governments in many countries. It is provided to improve safety for pedestrians and increase visibility for motorists. It also provides a more inviting environment after dark.

(b) With reference to the data above and your knowledge of economics, analyse why the free rider problem means street lighting is provided by the government.

(6)

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# 1. Question 4 · part b · 6 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part b only.

The English city of Kingston upon Hull has an independent telecoms network known as Kingston Communications (KCOM). It is the only provider serving the city and its surrounding towns and villages. People living there do not have many broadband options. The network remained independent when most others joined together to become British Telecom (BT) and it has remained as a separate company to this day.

As well as broadband, KCOM also offers home phone and mobile deals which are available at a discounted price when purchased together. It will soon offer high-quality television packages. Its broadband is fast, with fibre optic packages which can reach impressive speeds of up to 250 Mb. However, these are not available everywhere in the region and KCOM's packages are more expensive than those in the rest of the country. This is especially true as they have low download limits.

(Source adapted from: https://www.broadbandchoices.co.uk/guides/broadband/hull-broadband)

(c) With reference to the data above and your knowledge of economics, evaluate whether a monopoly such as KCOM is always bad for the consumer.

(12)

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# 1. Question 4 · part b · 6 marks · Working and answer

Working and answer

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## 2. Question 2 · part d · 1 mark

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

**2** (a) Which **one** of the following might be an aim of a firm in the private sector?

(1)

- ☒ **A** Maximise costs
- ☒ **B** Reduce interest rates
- ☒ **C** Maximise profits
- ☒ **D** Reduce economies of scale

Cyprus has a mixed economy.

(b) Which **one** of the following is most likely to be provided by the public sector in Cyprus?

(1)

- ☒ **A** Cars
- ☒ **B** Roads
- ☒ **C** Driving lessons
- ☒ **D** Petrol

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## 2. Question 2 · part d · 1 mark

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

France is a developed country.

(c) On the blank pie chart below, draw and label the approximate sizes of the primary (P), secondary (S) and tertiary (T) sectors for a developed country such as France.

(3)

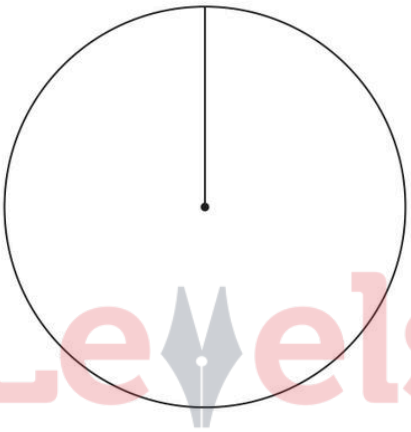


**Figure 2**

(d) Define the term free rider.

(1)

(e) Describe **one** reason why congestion is an example of an external cost.

(2)

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## 2. Question 2 · part d · 1 mark · Working and answer

Working and answer

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# 3. Question 2 · part g · 3 marks

2020 JAN · Paper 01

Further practice · Attempt part g only.

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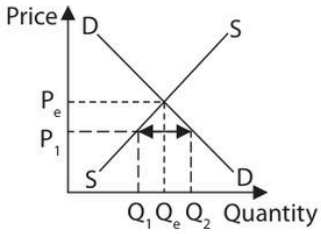
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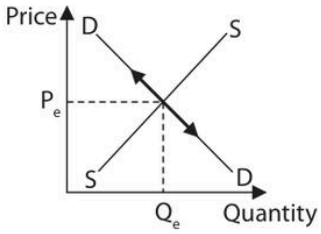
**2** (a) Which **one** of the following diagrams shows excess demand?

**(1)**

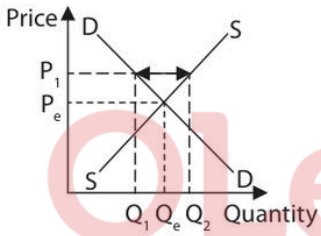
☒ **A**



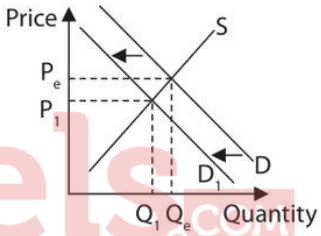

☒ **B**




☒ **C**




☒ **D**




(b) Land can be described as which **one** of the following?

**(1)**

☒ **A** A diseconomy of scale

☒ **B** A factor of production

☒ **C** A variable cost

☒ **D** A market failure

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## 3. Question 2 · part g · 3 marks

2020 JAN · Paper 01

Further practice · Attempt part g only.

(c) State the formula for profit.

(1)

(d) Calculate the **price elasticity of supply (PES)** for a product when price increases by 30% and quantity supplied increases by 36%. You are advised to show your working.

(2)

(e) Define the term finance.

(1)

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# 3. Question 2 · part g · 3 marks

2020 JAN · Paper 01

Further practice · Attempt part g only.

The Irish Artisan Charcoal Company (IACC) is the only firm producing charcoal in Ireland. Damaged trees which would otherwise have been left to rot are used to produce top quality charcoal.

(Source adapted from: https://www.limerickleader.ie/video/home/348948/watch-irish-artisan-charcoal-company-in-limerick-blazing-a-trail-with-unique-product.html)

(f) Describe the type of market structure in which the Irish Artisan Charcoal Company operates.

(2)

Street lighting is considered to be a public good.

(g) Explain **one** reason why this may cause a free rider problem.

(3)

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# 3. Question 2 · part g · 3 marks

2020 JAN · Paper 01

Further practice · Attempt part g only.

Wonderful Wax is a small, UK-based firm that produces candles. The candles are environmentally friendly because they are handmade with soy wax and pure essential oils. This is instead of harmful paraffin wax which produces toxins when burned. Elissa Newham set up the business because she wanted to help the environment. She was surprised when a larger firm made contact, offering its facilities to increase production if Elissa was prepared to share the profits.

(h) With reference to the data above and your knowledge of economics, assess whether Wonderful Wax would have more benefits from remaining a small firm rather than becoming a large firm.

(9)

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## 3. Question 2 · part g · 3 marks · Working and answer

Working and answer

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# 4. Question 2 · part c · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part c only.

**2** (a) Which **one** of the following products has a price inelastic demand?

**(1)**

| | Product | Price elasticity of demand (PED) |
|---|---|---|
| ☒ **A** | **W** | –1.5 |
| ☒ **B** | **X** | –1 |
| ☒ **C** | **Y** | –0.5 |
| ☒ **D** | **Z** | 1 |

(b) Which **one** of the following is the formula for calculating average cost?

**(1)**

☒ **A** $\frac{\text{Total fixed cost}}{\text{Quantity produced}}$

☒ **B** $\frac{\text{Total revenue}}{\text{Quantity produced}}$

☒ **C** $\frac{\text{Total cost}}{\text{Quantity produced}}$

☒ **D** $\frac{\text{Total variable cost}}{\text{Quantity produced}}$

(c) State **one** reason why governments intervene in the economy.

**(1)**

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# 4. Question 2 · part c · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part c only.

Figure 2 shows selected financial data of a firm selling shoes.

| | Cost/Revenue per pair of shoes ($) |
|---|---|
| Selling price | 89 |
| Raw materials | 17 |
| Labour | 35 |

**Figure 2**

(d) Calculate the **profit or loss** for the firm for each pair of shoes. You are advised to show your working.

(2)

Governments can use fines to deal with externalities.

(e) Define the term fine.

(1)

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## 4. Question 2 · part c · 1 mark · Working and answer

Working and answer

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## 5. Question 3 · part a · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part a only.

**3** (a) Which **one** of the following is likely to lead to an inefficient allocation of resources?

**(1)**

- ☒ **A** Migration
- ☒ **B** Competition
- ☒ **C** Lack of information
- ☒ **D** Division of labour

(b) Which **one** of the following is an example of an occupation in the tertiary sector?

**(1)**

- ☒ **A** Sheep farmer
- ☒ **B** Factory worker
- ☒ **C** Coal miner
- ☒ **D** Sales assistant

(c) Using the diagram below, draw the effects of a minimum wage ($W_1$) that has been set above the equilibrium wage ($W_e$). Label the new quantity of labour demanded and the new quantity of labour supplied.

**(3)**

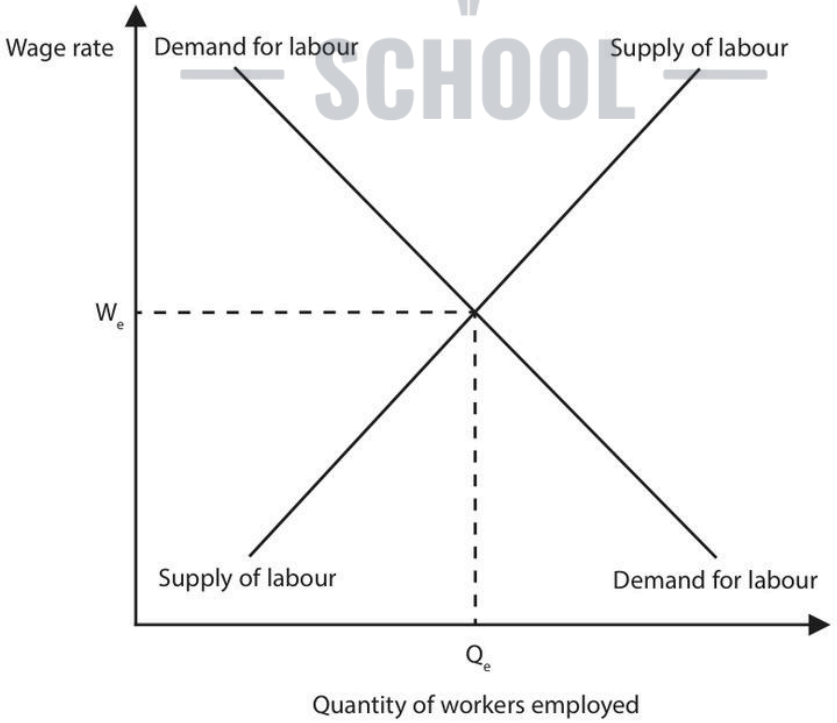



**Figure 3**

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## 5. Question 3 · part a · 1 mark · Working and answer

Working and answer

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## 6. Question 3 · part d · 6 marks

2023 JAN · Paper 01

Further practice · Attempt part d only.

**3** (a) Which **one** of the following could be described as an economic need?

**(1)**

- ☒ **A** Haircut
- ☒ **B** Holiday
- ☒ **C** Shelter
- ☒ **D** Smartphone

(b) Which **one** of the following is an example of an external benefit?

**(1)**

- ☒ **A** Education
- ☒ **B** Air pollution
- ☒ **C** Overcrowding
- ☒ **D** Resource depletion

(c) Draw a supply curve on the diagram below that has price elastic supply (PES). Label the impact of a price change from $P_1$ to $P_2$ and its impact on quantity from $Q_1$ to $Q_2$.

**(3)**




**Figure 3**

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## 6. Question 3 · part d · 6 marks

2023 JAN · Paper 01

Further practice · Attempt part d only.

Every year in the UK, heavy rainfall is a cause of severe flooding in some areas. Calderdale in Yorkshire has been badly affected for a number of years. Increased water levels from the River Calder caused damage to residential properties, business premises and roads. The UK Government provided a £32m flood defence scheme to try to reduce the problem.

(Source adapted from: https://www.halifaxcourier.co.uk/news/politics/ps44m-to-be-spent-on-flood-defences-in-hebden-bridge-3327606)

(d) With reference to the data above and your knowledge of economics, analyse why the UK Government might provide flood defences.

(6)

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## 6. Question 3 · part d · 6 marks

2023 JAN · Paper 01

Further practice · Attempt part d only.



# 6. Question 3 · part d · 6 marks

2023 JAN · Paper 01

Further practice · Attempt part d only.

In 2021, Morrisons, a leading UK-based supermarket chain, became the first in the retail industry to pay workers at least £10 per hour. This increased its hourly wage rate from £9.20 to at least £10 per hour for its 110,000 workers. This increase was a result of negotiations with the trade union, Usdaw. The majority of Morrisons' workers are trade union members.

(Source adapted from: https://www.hrmagazine.co.uk/content/news/morrisons-is-the-first-uk-supermarket-to-pay-minimum-10-an-hour)

(e) With reference to the data above and your knowledge of economics, assess the impact of trade union activity on Morrisons and its employees.

(9)

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## 6. Question 3 · part d · 6 marks · Working and answer

Working and answer

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## 7. Question 4 · part b · 6 marks

2023 JUN · Paper 01

Further practice · Attempt part b only.

**4** A sports arena in Guatemala holds a maximum of 20,000 people for each event. The quantity of tickets demanded for the most recent events is shown below.

| Event | Quantity of tickets demanded |
|---|---|
| A | 22,200 |
| B | 21,000 |
| C | 20,250 |

**Figure 5**

(a) Calculate the **excess demand** for tickets if the quantity supplied for all three events totals 60,000. You are advised to show your working.

**(2)**

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## 7. Question 4 · part b · 6 marks

2023 JUN · Paper 01

Further practice · Attempt part b only.

Zambia has a defence force that includes both an army and an air force. The Zambian Government is responsible for the provision of this defence force.

(b) With reference to the data above and your knowledge of economics, analyse why the free rider problem has led to the Zambian Government providing the defence force.

**(6)**

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## 7. Question 4 · part b · 6 marks

2023 JUN · Paper 01

Further practice · Attempt part b only.



## 7. Question 4 · part b · 6 marks

2023 JUN · Paper 01

Further practice · Attempt part b only.

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Polluted air is said to cause over 1,000 deaths each year in Greater Manchester, UK. To reduce the problem, an 800 km$^2$ 'Clean Air Zone' (CAZ) has been proposed, where drivers of certain vehicles would have to pay between £7.50 and £60.00 each day they drive on roads in the region.

All lorries, buses, taxis and vans that do not meet the emission standards will be affected. It is hoped the tax will encourage vehicle owners to drive vehicles that are more environmentally friendly.

However, small business owners in particular will be affected by the proposals. An ice-cream business, established over 100 years ago, stated costs would increase by £29 000 a year for its vans to use roads in the region. It cannot afford to replace them and like many other firms risks going out of business.

(c) With reference to the data above and your knowledge of economics, evaluate the effectiveness of government regulations in dealing with externalities such as air pollution in Greater Manchester.

**(12)**

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## 7. Question 4 · part b · 6 marks · Working and answer

Working and answer

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# 8. Question 1 · part i · 6 marks

2023 NOV · Paper 01

Further practice · Attempt part i only.

**1** (a) A trade union is an organisation that aims to improve

(1)

- ☒ **A** profit margins
- ☒ **B** tax revenue
- ☒ **C** the environment
- ☒ **D** working conditions

(b) Figure 1 below shows a change in demand for tickets to a sporting event.

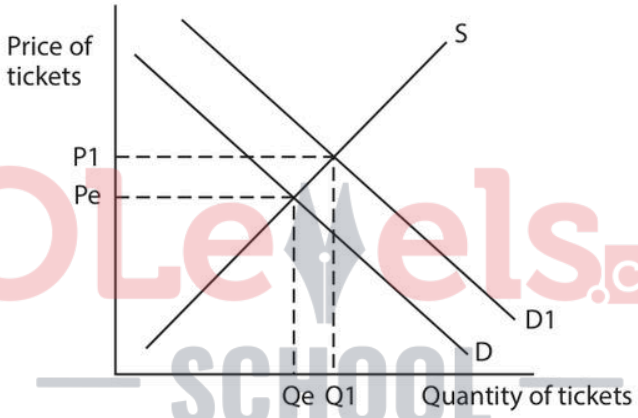



**Figure 1**

Which **one** of the following statements is true?

(1)

- ☒ **A** When the price falls, the quantity demanded falls
- ☒ **B** The demand curve slopes upwards from left to right
- ☒ **C** An increase in demand is shown by a shift to the right
- ☒ **D** A decrease in demand is shown by a shift to the right

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# 8. Question 1 · part i · 6 marks

2023 NOV · Paper 01

Further practice · Attempt part i only.

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(c) What is meant by the term economies of scale?

(2)

(d) State **one** main feature of an oligopoly.

(1)

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(e) Define the term excess supply.

(1)

# 8. Question 1 · part i · 6 marks

2023 NOV · Paper 01

Further practice · Attempt part i only.

A firm increases supply by 4.3% following a price increase of 3.7%.

(f) Calculate, to two decimal places, the **price elasticity of supply (PES)** for the firm. You are advised to show your working.

**(2)**

Figure 2 shows the market for electric cars.

(g) Using the diagram below, draw the likely effects on the market for electric cars following the introduction of a subsidy on electric cars. Label the new curve, the new equilibrium price and the new equilibrium quantity.

**(3)**

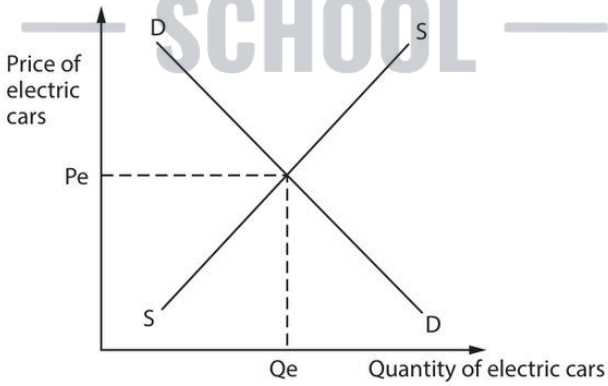



**Figure 2**

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# 8. Question 1 · part i · 6 marks

2023 NOV · Paper 01

Further practice · Attempt part i only.

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The Brazilian Government announced plans to privatise one of its state-owned food companies.

(h) Explain **one** possible advantage of privatisation for consumers in Brazil.

**(3)**

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# 8. Question 1 · part i · 6 marks

2023 NOV · Paper 01

Further practice · Attempt part i only.

The Tallinn Song Festival Grounds (TSFG) is a popular live music venue in Estonia and can hold 65,000 people. Although it is very popular there are only a few firms selling food and drink at the TSFG venue.

The lack of competition at the TSFG venue means that service is slow and queues are long. Food and drink prices are expensive and security checks mean that no food and drink can be brought in from outside the TSFG venue.

(i) With reference to the data above and your knowledge of economics, analyse how market failure is likely to occur at the TSFG venue.

**(6)**

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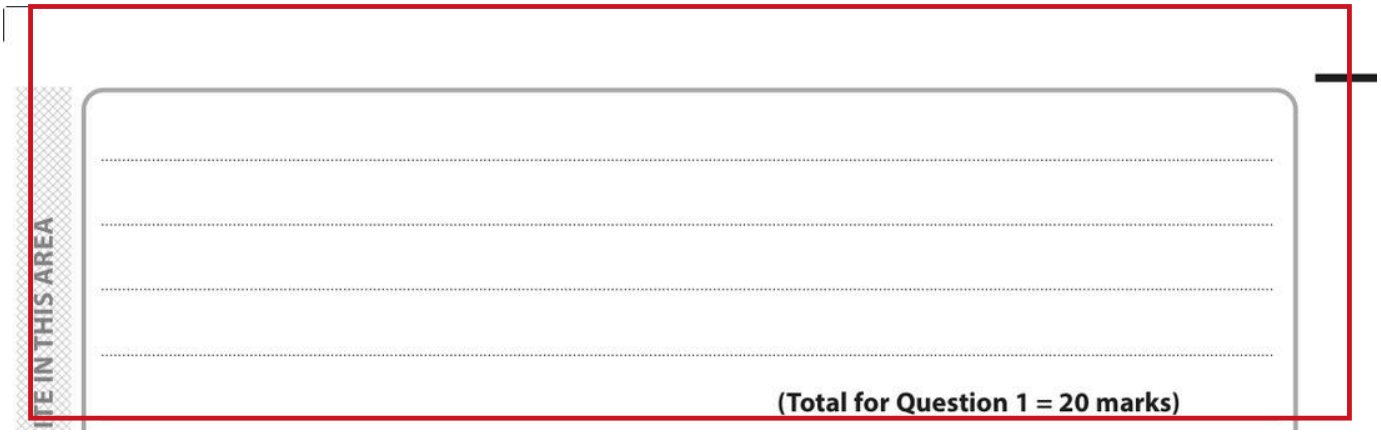
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## 8. Question 1 · part i · 6 marks

2023 NOV · Paper 01

Further practice · Attempt part i only.




Working and answer

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## 9. Question 4 · part b · 6 marks

2024 NOV · Paper 01

Further practice · Attempt part b only.

**4** A factory has been unable to meet the demand for its product due to a shortage of raw materials. The quantity demanded for the product during each of the first three months of 2024 is shown in Figure 5.

| Month | Quantity of products demanded |
|---|---|
| January | 13,600 |
| February | 14,200 |
| March | 11,700 |

**Figure 5**

(a) Calculate the **excess demand** for the product for the first three months of 2024, if the quantity supplied during each month was 11,000 (33,000 in total). You are advised to show your working.

**(2)**

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## 9. Question 4 · part b · 6 marks

2024 NOV · Paper 01

Further practice · Attempt part b only.

Hiking is a popular activity for visitors to the Caribbean island of Saint Vincent. Tourists can hire a guide to accompany them on the popular route of La Soufrière Volcano by booking online before arriving on the island and the price is $34. However, the price is only $20 if the guide is booked directly on arrival. Some tourists have complained because they did not know this was possible.

(b) With reference to the data above and your knowledge of economics, analyse a possible reason why the market for tourism services is an inefficient allocation of resources.

(6)

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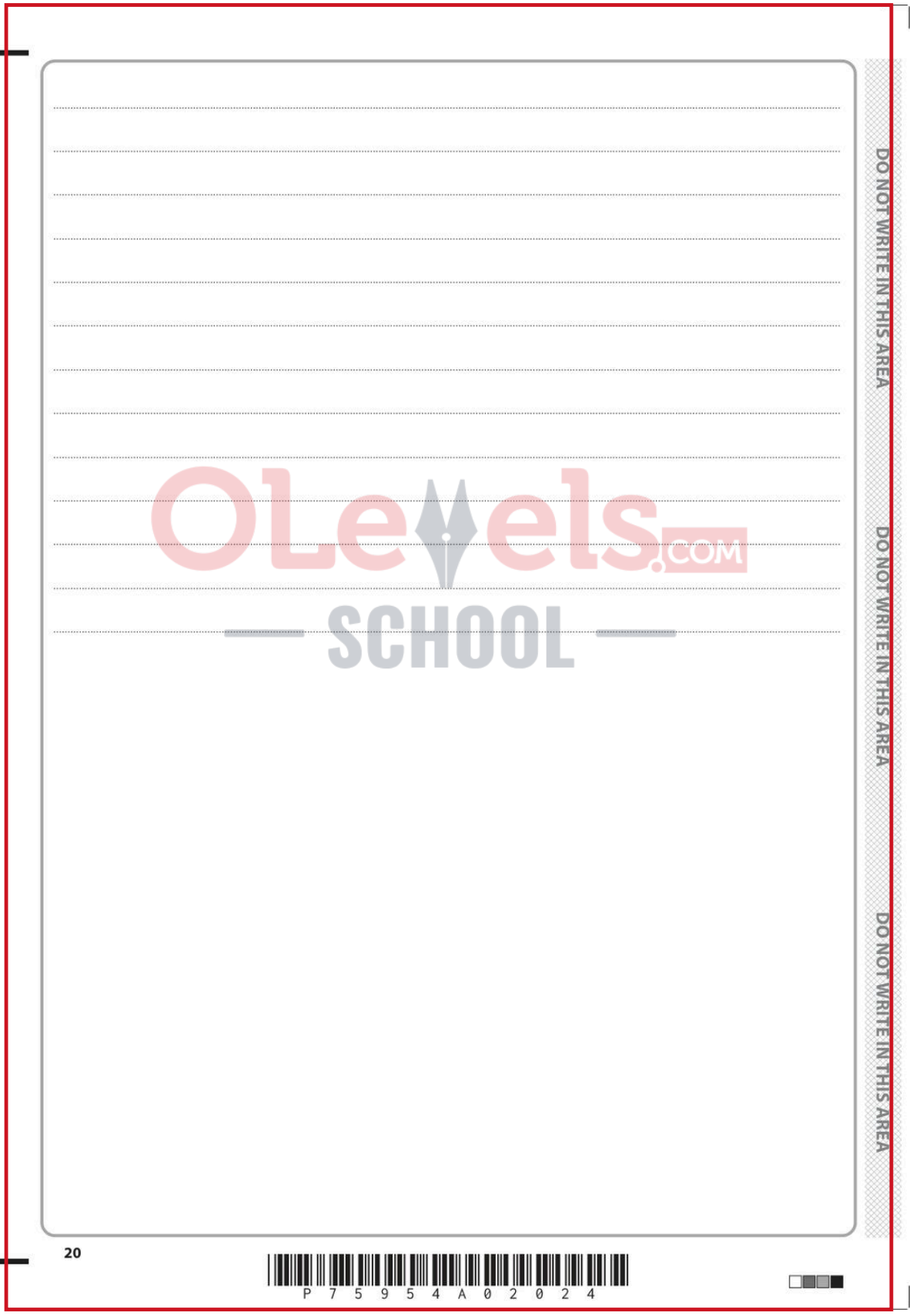
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## 9. Question 4 · part b · 6 marks

2024 NOV · Paper 01

Further practice · Attempt part b only.



# 9. Question 4 · part b · 6 marks

2024 NOV · Paper 01

Further practice · Attempt part b only.

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Twitter, one of the most popular social media platforms, was bought for $44bn by entrepreneur Elon Musk in 2022.

The aims of Elon Musk include the creation of an 'everything app' – a platform that includes messaging, payments, ridesharing, food delivery and more. To support this, Elon Musk introduced major changes to Twitter, including renaming it 'X'.

The number of X users increased to a new high of 530 million by 2023, following an increase in publicity about the change of name. This meant X did not need to pay high advertising costs to promote the platform.

Advanced technology will be introduced by X to enable users to operate multiple functions on the new platform. This research and development is likely to lead to a significant cost for the firm.

The platform has experienced a decrease of 65% in advertising revenue. Thousands of workers have already lost their jobs, including those in managerial roles.

(c) With reference to the data above and your knowledge of economics, evaluate whether X is likely to benefit from economies of scale.

(12)

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## 9. Question 4 · part b · 6 marks · Working and answer

Working and answer

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## 10. Question 3 · part a · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part a only.

**3** (a) A free rider can be defined as an individual who does which **one** of the following?

**(1)**

- ☒ **A** Enjoys the benefit of a good after paying for it
- ☒ **B** Enjoys the benefit of a good but does not pay for it
- ☒ **C** Gives a good away after paying for it
- ☒ **D** Neither uses a good nor pays for it

(b) Which **one** of the following is an example of an occupation in the tertiary sector?

**(1)**

- ☒ **A** House builder
- ☒ **B** Cattle farmer
- ☒ **C** Hotel receptionist
- ☒ **D** Forestry worker

(c) Using the diagram below, draw the effect of a minimum wage ($W_1$) that has been set above the equilibrium wage ($W_e$). Label the new quantity of labour demanded and the new quantity of labour supplied.

**(3)**

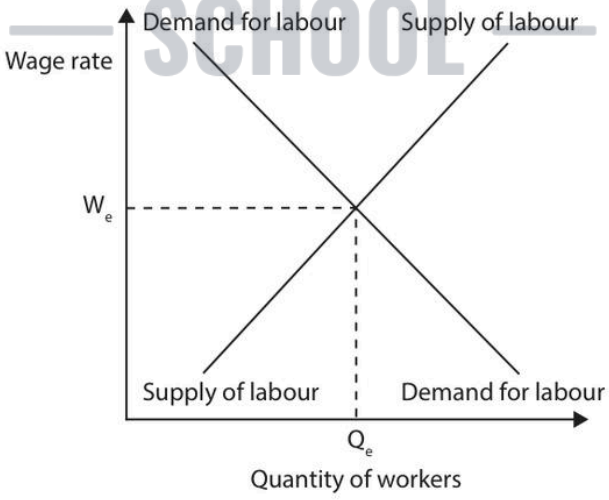



**Figure 3**

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P 7 2 1 3 2 A 0 1 2 2 4

## 10. Question 3 · part a · 1 mark · Working and answer

Working and answer

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